JOSEPH JIN

josephjin@berkeley.edu | linkedin.com/in/joseph-jin | (512)565-0000

Experience

Software Engineering, Databricks **July 2024 - Present**

- Architected and implemented high-availability billing service for new LLM batch inference product, capable of processing 1,000+ records/second with 99.95%+ uptime using Scala, MySQL, Kafka, Apache Spark, and Kubernetes operators
- Reduced complexity of ML inference cluster by rolling out new Kubernetes resources for autoscaling, creating Python jobs to monitor customer workloads, writing Kubernetes automations for unblocking failed rollouts, and patching user API routing mechanics in Scala
- Implemented and rolled out tools for ML inference Docker images to enforce and monitor network policies with Scala, Prometheus, and Shell scripts, enabling 50+ enterprise customers with compliance requirements on egress control to use Databricks inference
- Led critical incident response as on-call engineer, resolved multiple high-priority service outages, interacted with 10+ customers for troubleshooting and customizing compute capacity, maintained 99.5% service uptime

Software Engineering Intern, Databricks **May 2023 - August 2023**

- Designed and implemented Kubernetes controllers to speed up infrastructure rollouts on ML-inference compute clusters hosting 2,000+ customer deployments, reducing manual labor for each rollout by 10 hours, using Go and Kubernetes controller-runtime SDK
- Developed new deployment scheme for special small ML-serving types, ensuring zero downtime for 500+ additional enterprise customers during infrastructure changes, also using custom Kubernetes controllers

Researcher, Berkeley Artificial Intelligence Research Lab **February 2022 - December 2022**

- Worked with Professor Trevor Darrell and PhD candidate Zhuang Liu to show how accuracy and convergence of vision models can be improved by scheduling regularization strength during training (accepted to ICML 2023 arxiv.org/pdf/2303.01500.pdf)
- Designed and executed experiments, implemented training techniques for commonly used vision models with PyTorch and timm, developed tools for benchmarking, analysis, and loss-landscape visualizations with Pandas

Course Staff, UC Berkeley EECS Department **August 2022 - December 2022**

- Taught subjects such as vector calculus, PCA, SVD, convex optimization, and duality in office hours for EECS 127 (Optimization Models) ; answered questions on class forums; graded assignments; debugged projects and tests; proctored exams

Software Development Intern, Amazon **May 2022 - August 2022**

- Developed low-latency pipeline reducing time required to monitor errors made by fraud detection ML models from 1 hour per 20,000 samples to under 5 seconds using Java, Typescript, Dagger, and AWS CDK
- Programmed and deployed cloud infrastructure for ML engineers in Amazon Catalogs to speed up data storage, querying, cleaning, and metric creation using AWS CloudFormation, S3, Lambda, DynamoDB, CloudWatch, and QuickSight

Data Consultant, Tribe Capital **February 2022 - May 2022**

- Developed end-to-end ML labeling, training and prediction pipeline for Tribe Capital (venture capital firm) to identify prospective startups; trained ensemble learning models using Pandas, Sklearn, and NumPy, achieving a greater than 90% prediction accuracy

Product Manager, Stroll **April 2021 - December 2021**

- Directed product development of early-stage startup enrolled in SkyDeck Batch 13 incubator program
- Led team of five engineers using the Agile workflow to develop frontend, backend and algorithms for safety based GPS navigation app
- Deployed beta onto iOS and Android for the city of Berkeley, acquiring 500+ active users during one week of pilot testing

Education

University of California, Berkeley **Class of 2024**

B.S. Electrical Engineering and Computer Sciences GPA: 3.9/4.0

Relevant Coursework

Machine Learning, Reinforcement Learning, Artificial Intelligence, Deep Learning, Computer Vision, Advanced Algorithms, Data Structures, Optimization Models, Operating Systems, Database Systems, Quantum Computing, Computer Architecture, Computer Security, Cryptography, Discrete Math, Probability, Structure of Computer Programs, Circuit Design and Theory I/II, Physics for Engineers

Skills

Languages: C, Python, Scala, Java, Golang, Swift, Javascript, R, Matlab

Languages/Tools: PyTorch, Scikit-learn, NumPy, Pandas, Kubernetes, Knative, Docker, CVXPY, Git, AWS, MySQL, MongoDB

Frontend/Backend: React.js, React Native, Vue.js, Node.js, Apple UIKit, Selenium

- **Eta Kappa Nu Honors Society** - top 20% of Berkeley EECS undergraduate students by GPA
- **Gold Division, USA Computing Olympiad -** top 6% of competitors in the United States